and making true international partnerships, the justifications become more compelling.

Until a national debate on this topic has transpired, until we have a better idea of the rationale and the cost/benefit ratio of human missions to Mars, what should we do? My suggestion is that we pursue research and development projects that can be justified on their own merits or by their relevance to other goals, but that can also contribute to human missions to Mars should we later decide to go. Such an agenda would include:

- U.S. astronauts on the Russian space station *Mir* for joint flights of gradually increasing duration, aiming at one to two years, the Mars flight time.
- Configuration of the international space station so its principal function is to study the long-term effects of the space environment on humans.
- Early implementation of a rotating or tethered "artificial gravity" module on the international space station, for other animals and then for humans.
- Enhanced studies of the Sun, including a distributed set of robot probes in orbit about the Sun, to monitor solar activity and give the earliest possible warning to astronauts of hazardous "solar flares"—mass ejections of electrons and protons from the Sun's corona.
- U.S./Russian and multilateral development of *Energiya* and *Proton* rocket technology for the U.S. and international space programs. Although the United States is unlikely to depend primarily on a Soviet booster, *Energiya* has roughly the lift of the *Saturn V* that sent the *Apollo* astronauts to the Moon. The United States let the *Saturn V* assembly line die, and it cannot readily be resuscitated. *Proton* is the most reliable large booster now in service. Russia is eager to sell this technology for hard currency.
- Joint projects with NASDA (the Japanese space agency) and Tokyo University, the European Space Agency, and the Russian Space Agency, along with Canada and other nations. In most